# Two- and three-point objects

Plain lines expose connectivity. IDs identify recipes; rotated layouts are not extra graph topologies.

### I01 Two-point object

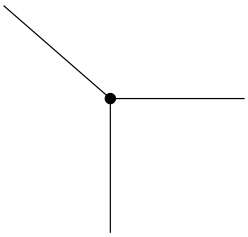
`\TwoPointCorr`



Two external segments meet at one marked point.
A central point or optional blob, not a loop correction.

### I02 Three-point: upper left

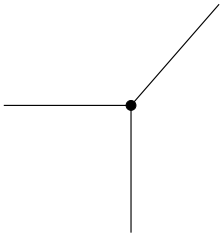
`\ThreePointCorr`



One trivalent vertex; logical slot 1 points upper left.
Four orientations retain the same three logical slots.

### I03 Three-point: upper right

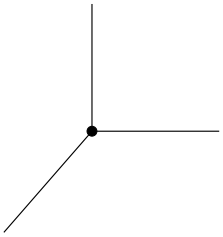
`\ThreePointCorr`



One trivalent vertex; logical slot 1 points upper right.
Four orientations retain the same three logical slots.

### I04 Three-point: lower left

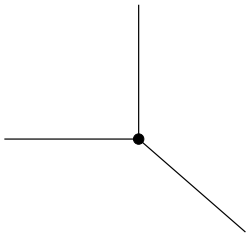
`\ThreePointCorr`



One trivalent vertex; logical slot 1 points lower left.
Four orientations retain the same three logical slots.

### I05 Three-point: lower right

`\ThreePointCorr`



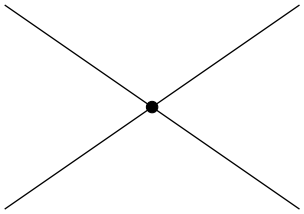
One trivalent vertex; logical slot 1 points lower right.
Four orientations retain the same three logical slots.

# Contact and pairing layouts

Pairing presets are shown as implemented. A missing bridge is meaningful, not an omitted annotation.

### I06 Central contact

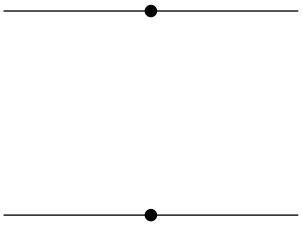
`\FourPointCorr`



Four external edges meet at one quartic vertex.
Connected four-point contact.

### I07 Horizontal pairing

`\FourPointCorr`



Top pair and bottom pair each meet separately.
Disconnected pairs; there is no vertical exchange edge.

### I08 Vertical pairing

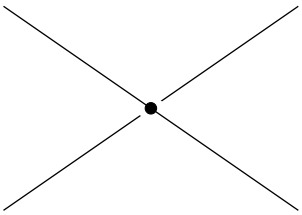
`\FourPointCorr`



Left pair and right pair each meet separately.
Disconnected pairs; there is no horizontal exchange edge.

### I09 Crossed contact layout

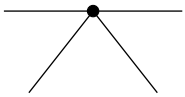
`\FourPointCorr`



Curved and straight external edges meet at a central point.
The central dot is an interaction, not an over/under crossing.

### I10 Flat contact

`\FlatContactCorr`



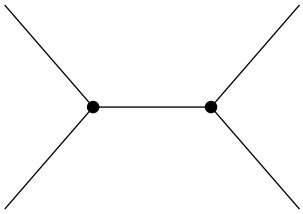
One quartic vertex with a flat upper pair.
A contact layout, not an additional loop topology.

# Tree exchanges and four-legged bubbles

External slot order: 1 upper left, 2 lower left, 3 upper right, 4 lower right. No momentum annotations.

### I11 s-channel tree

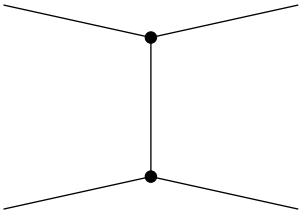
`\STreeChannelCorr`



Two trivalent vertices joined by one exchange edge.
Insertion support is narrower than ordinary tree support.

### I12 t-channel tree

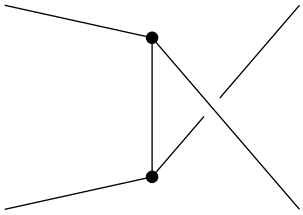
`\TTreeChannelCorr`



Two trivalent vertices joined by one exchange edge.
Insertion support is narrower than ordinary tree support.

### I13 u-channel tree

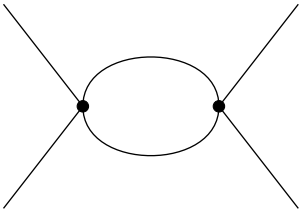
`\UTreeChannelCorr`



Two trivalent vertices joined by one exchange edge.
Insertion support is narrower than ordinary tree support.

### I14 s-channel bubble

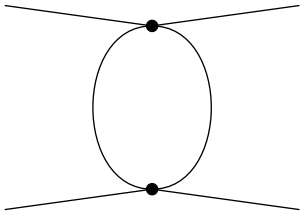
`\SChannelCorr`



Two quartic vertices joined by two internal edges.
Four-legged bubble already exists; unlike a self-energy insertion.

### I15 t-channel bubble

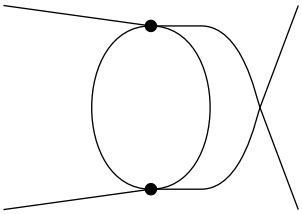
`\TChannelCorr`



Two quartic vertices joined by two internal edges.
Four-legged bubble already exists; unlike a self-energy insertion.

### I16 u-channel bubble

`\UChannelCorr`



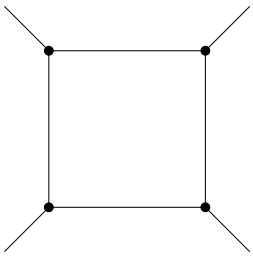
Two quartic vertices joined by two internal edges.
Four-legged bubble already exists; unlike a self-energy insertion.

# Boxes and half-box layouts

The ordinary box already defaults to a square. Half-box presets include identity-specific marker variants.

### I17 Square box

`\BoxLoopCorr`

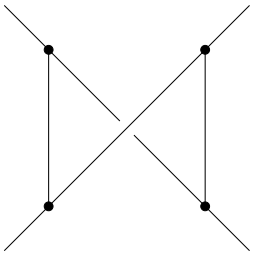


Four trivalent vertices around a single loop.

Default inner spans are both 1.34; explicit overrides remain possible.

### I18 Existing crossed box

`\CrossBoxCorr`



Fixed left/right edges plus two crossing diagonals.

No interaction at crossing; internal momentum/index pass is incomplete.

### I19 Straight half-box

`\HalfBoxCorr`

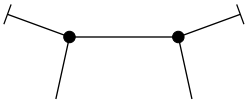


Two trivalent vertices with a horizontal bridge.

An open tree helper, despite the half-box name.

### I20 Angled half-propagator

`\HalfBoxCorr`

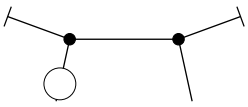


Angled upper stubs on the same two-vertex graph.

A layout variant of the half-box, not a new loop.

### I21 Left half-propagator marker

`\HalfBoxCorr`

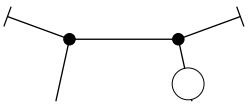


Angled half-box with the left lower-leg circle preset.

Blank circle retained to show the preset; same underlying graph.

### I22 Right half-propagator marker

`\HalfBoxCorr`



Angled half-box with the right lower-leg circle preset.

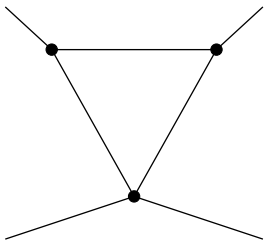
Blank circle retained to show the preset; same underlying graph.

# Four triangle-contact orientations

Two trivalent vertices and one quartic vertex. These orientations already exist; label independence remains future work.

### I23 Quartic contact: bottom

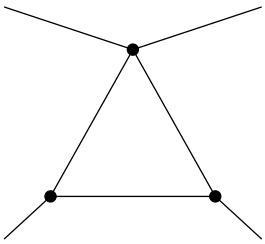
`\TriangleContactCorr`



The quartic/contact vertex lies on the bottom side.
Not the regular three-trivalent-vertex triangle attached to a tree.

### I24 Quartic contact: top

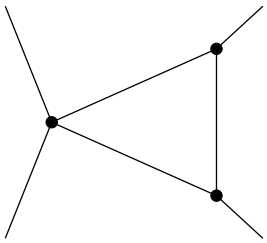
`\TriangleContactCorr`



The quartic/contact vertex lies on the top side.
Not the regular three-trivalent-vertex triangle attached to a tree.

### I25 Quartic contact: left

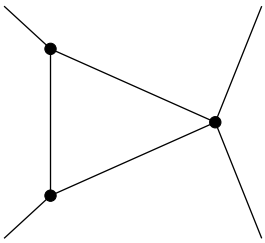
`\TriangleContactCorr`



The quartic/contact vertex lies on the left side.
Not the regular three-trivalent-vertex triangle attached to a tree.

### I26 Quartic contact: right

`\TriangleContactCorr`



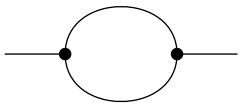
The quartic/contact vertex lies on the right side.
Not the regular three-trivalent-vertex triangle attached to a tree.

# Loops and the insertion prototype

Only the insertion pictured here is currently supported. Unsupported insertion settings can be silently ignored.

### I27 Standalone self-energy bubble

`\BubbleCorr`

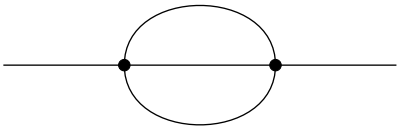


Two trivalent vertices and two internal arcs; one loop.

Two external legs; distinct from a four-legged channel bubble.

### I28 Sunset

`\SunsetDiagram`

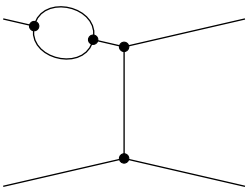


Two quartic vertices joined by three internal edges.

Two loops, not a one-loop self-energy bubble.

### I29 Bubble on t-tree leg 1

`\TTreeChannelCorr`



A self-energy bubble splits the upper-left external leg.

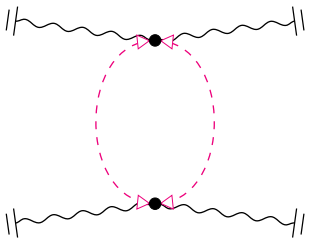
Only t-tree external slot 1 is implemented; other slots are no-ops.

# Existing styles and annotations

Representative combinations, not a promise of every style on every topology. Recipes include all displayed options.

## A01 Your pink Proca t-bubble

`\TChannelCorr`

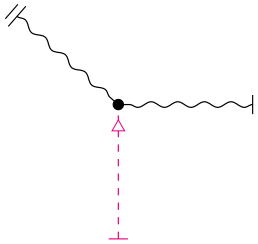


Four double-capped EW legs and two pink Proca internals.

M01 verifies automatic contact; custom legacy distances retain their tuning.

## A02 Polarization and mixed legs

`\ThreePointCorr`

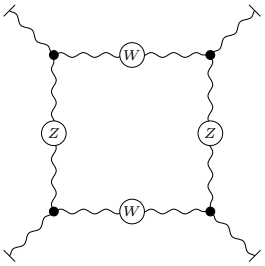


Double cap, single cap, and a pink scalar-polarized leg.

Scalar polarization is a style, not another topology.

## A03 EW circles and single caps

`\BoxLoopCorr`

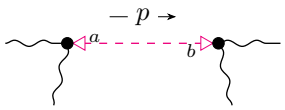



Independent internal particle circles and external endcaps.

Species labels decorate the existing square loop.

## A04 Bridge momentum and indices

`\HalfBoxCorr`

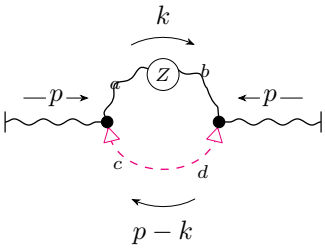



A pink bridge with one momentum and endpoint indices.

Omit global line=plain: it overrides the bridge-specific style.

## A05 Curved momentum and indices

`\BubbleCorr`

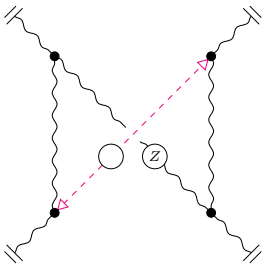



External flow, curved loop momenta, and four endpoint indices.

Representative annotation layout; crowded cases need dedicated checks.

## A06 Crossing and Proca circles

`\CrossBoxCorr`




A blank circle on one pink diagonal and a Z on the other.

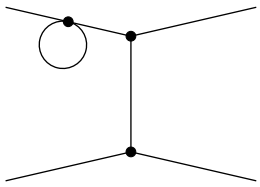
Crossing stays unconnected; no internal momenta or indices claimed.

# Seagull insertion examples

Rendered from codex/seagull-insertion at 5a657cd. Not yet merged into main or the current checkout. Only t-tree external slot 1 is supported.

### U01 Seagull: bare structure

`\TTreeChannelCorr`

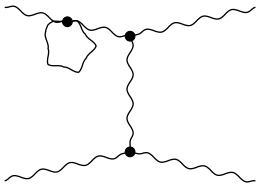


One quartic point on an external leg carries a self-loop.

Unmerged prototype: t-tree leg 1 only; all other placements remain future work.

### U02 Seagull: EW line style

`\TTreeChannelCorr`



The same supported graph with wavy EW propagators.

No caps, circles, labels, or momenta; requires the unmerged branch.